(1.) Course Title: INFPHD032 Type theory	**Credits: 6**
Course **Classification: Optional**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: 100% theoretical (credit%)	
Type of Class: lecture and **number of hours**: **2/week** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: -	
Assessment Method (Exam / Midterm Grade / **Other**[3]): **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: offered grade based on giving a seminar on a topic related to type theory	
Course Placement in Curriculum (Semester Number): -	
Prerequisites (*if any*): none, but MSc courses in type theory and language type systems are recommended	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

Aim of the subject: Understanding Martin-Löf's type theory through the usage of the Agda proof assistant. Basic metatheoretic results concerning type theory using the methods of category theory. Discussion of the new research directions and new results in type theory.

At the research seminar we introduce the syntax and semantics of Martin-Löf's type theory. We study it using the abstract language of category theory and using the implementation Agda. We discuss the following topics:

* basic category theory

* Categories withFamilies (CwF), as a notion of model of type theory

* the syntax as the initial model

* inductive types and higher inductive types

* coinduction

* the equality type

* intensional type theory, extensional type theory, homotopy type theory

* models of type theory: set, graph, reflexvive graph, setoid, presheaf, simplicial, cubical

* parametricity, definable functions

* normalisation proof using logical predicates

2-5 Most Important Required and Recommended Literature
List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

- Bocquet-Kaposi-Kovács. Introduction to the metatheory of type theory video seminar series https://www.youtube.com/playlist?list=PLb_d4YUzq4lMgnFwBY4CJuHF911WWiocU
- Ambrus Kaposi. Second-order generalised algebraic theories, by examples. Draft course notes, 2025. https://akaposi.github.io/sogat-paris.pdf
- Univalent Foundations Project. Homotopy Type Theory –Univalent Foundations of Mathematics. Technical report, Institute for Advanced Study,2013.
- Philip Wadler. Programming Language Foundations in Agda
- Ambrus Kaposi. Bevezetés a homotópia-típuselméletbe. Csörnyei Zoltán (ed.) Tízéves az ELTE Eötvös József Collegium Informatikai Műhelye:2004-2014. ELTE Eötvös József Collegium, 2014. pp. 166-96
- Martin Hofmann. Syntax and Semantics of Dependent Types. In: Semantics and Logics of Computation, 1997, Cambridge University Press. pp. 79-130

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.[1]
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.

[1]**Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.
[2] e.g. case studies, role play, thematic presentations, etc.
[3] e.g. continuous debriefing, mid-year report
[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.

Course responsible (name, position, academic degree): **Dr. Ambrus Kaposi, Associate Professor, PhD**

Instructor(s) involved in teaching the course (if any) (name, position, academic degree):
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